

Masked diffusion model for generating discrete data

Reading material

- 1 Simple and Effective Masked Diffusion Language Models
.Subham Sekhar Sahoo,Marianne Arriola,Yair Schiff,Aaron Gokaslan,Edgar Marroquin,Justin T Chiu,Alexander Rush,Volodymyr Kuleshov. NeurIPS 2024
<https://s-sahoo.com/mdlm/>
- 2 Simplified and Generalized Masked Diffusion for Discrete Data.
Jiaxin Shi, Kehang Han, Zhe Wang, Arnaud Doucet, Michalis K. Titsias. NeurIPS 2024

Motivation

Useful for generating text, protein molecules from amino acid sequences, even images with discretized pixels

Default auto-regressive generation is slow, diffusion models allow for parallel generation of text. Examples

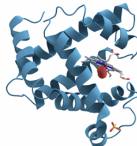
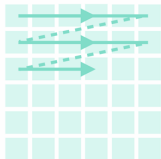
- 1 LLaDA: <https://ml-gsai.github.io/LLaDA-demo/>
- 2 Dream: <https://github.com/DreamLM/Dream>
- 3 Mercury coder: <https://www.inceptionlabs.ai/news>

Motivation

Fixed ordering not natural for some types of data¹.



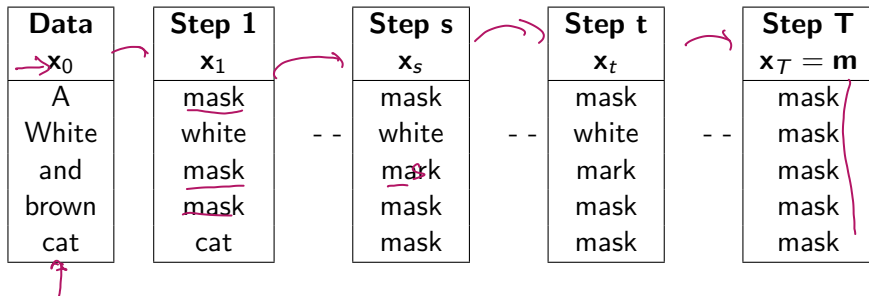
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DRETPVHAVNI INIWSAYKMTRYWCRIQCDSYWLWSGMTWRWCRWNREQPEWLSHDDMVQ  
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```



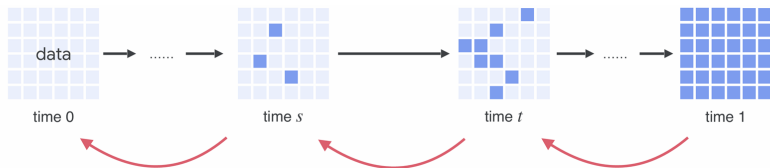
¹<https://neurips.cc/media/neurips-2024/Slides/93071.pdf>

Masked diffusion model for generating discrete data

Example (text)



Example: Discretized images



Notations I

Let X be a categorical random variable that can take one of R possible values from a vocabulary V . $|V| = R$

We denote its probability distribution $P(X)$ as a multinomial distribution with parameters $\pi \in \Delta^R$ that is $\pi_1 + \pi_2 + \dots + \pi_R = 1$ and $\pi_j \geq 0$

$$P(X) = \text{cat}(X; \pi = [\pi_1, \dots, \pi_R])$$

Example: $R = 3, V = [A, B, C]$

$$P(X) = \text{cat}(X; \pi = [\pi_1 = 0.1, \pi_2 = 0.5, \pi_3 = 0.4])$$

$$) P(X = B) = 0.5 = \pi_2$$

A categorical variable that takes value j , that is, $X = j$ can also be represented as a R dimension binary vector x that is 1 in exactly the j -th position and 0 everywhere else. We denote it as $B_R(j)$

Notations II

Example $X = 2$, $R = 4$ can be represented as $\mathbf{x} = [0, 1, 0, 0]$.

Consider a vocabulary $\mathcal{V}$ where the last token is a special mask token.

We use $\mathbf{m} = B_R(R)$ to denote the mask in binary form.

We will assume that the data to be generated is a single categorical variable $\mathbf{x}$ represented in the binary form that takes one of R possible values.

It will be easy to generalize to the case of generating multiple words since these are generated in parallel.

Forward process

During the forward process we start with an $\mathbf{x}_0 \in D$, calling it step 0.

$$q(\mathbf{x}_t | \mathbf{x}_{t-1})$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \text{cat}(\mathbf{x}_t; \pi = \alpha_t \mathbf{x}_0 + (1 - \alpha_t) \mathbf{m})$$

where $1 - \alpha_t$ is the probability of masking. We choose $\alpha_t < \alpha_s$ for $s < t$.

Another way of writing the above (assume $n = 1$)

$$\begin{aligned} q(x_t | x_0) &= 1 - \alpha_t \quad \text{if } x_t = \mathbf{m} \\ &= \alpha_t \quad \text{if } x_t = x_0 \\ &= 0 \quad \text{otherwise.} \end{aligned}$$

Example to clarify notation

$$V \equiv \{v_1, v_2 \dots v_R\}$$

$x_0 \equiv 0/1$ vector in

eg $x_0 = \underline{v_3}$

$$x_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$n = 1$ Generate one word.
 $v_R \equiv$ mask-token.
 R - dimensions.

$$m =$$

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$\underline{\alpha_t x_0 + (1 - \alpha_t) m} \equiv$$

$$\begin{bmatrix} 0 \\ 0 \\ \alpha_t \\ 0 \\ 0 \\ 1 - \alpha_t \end{bmatrix}$$

Sums to 1

we treat this vector as parameter of a multinomial distribution of R discrete values: which we denote as

$$x_t \sim \text{Cat}(x_t; \alpha_t x_0 + (1 - \alpha_t) m)$$

Forward process

$$\alpha_1 > \alpha_2 > \dots > \alpha_T \rightarrow 0$$

$$q(x_t | x_0)$$

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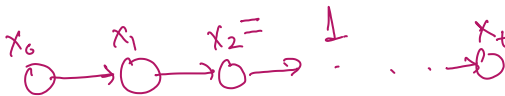
$$q(\underline{x}_t | \underline{x}_{t-1}) = \text{cat}(\underline{x}_t; \frac{\alpha_t}{\alpha_{t-1}} \underline{x}_{t-1} + (1 - \frac{\alpha_t}{\alpha_{t-1}}) \underline{m})$$

We choose α_t such that $\alpha_T \rightarrow 0$, so

$$\underline{q}(\underline{x}_T | \underline{x}_0) \rightarrow \underline{1} \text{ if } \underline{x}_T = \underline{m}$$

Elaborating the forward process.

$$\begin{aligned} \underline{q}(\underline{x}_t | \underline{x}_{t-1}) &= \frac{\alpha_t}{\alpha_{t-1}} \text{ if } \underline{x}_t = \underline{x}_{t-1} \text{ and } \underline{x}_{t-1} \neq \underline{m} \\ &= 1 - \frac{\alpha_t}{\alpha_{t-1}} \text{ if } \underline{x}_t = \underline{m} \text{ and } \underline{x}_{t-1} \neq \underline{m} \\ &\quad \text{if } \underline{x}_{t-1} = \underline{m} \text{ \& } \underline{x}_t = \underline{m} \end{aligned}$$



$$q(x_t | x_0) = q(x_t = x_0 | x_0)$$

$$\begin{aligned} x_t &\neq \text{mask} \\ x_t &= x_0 \end{aligned}$$

$$\underline{q(x_t, x_{t-1}, \dots, x_1 | x_0) = \prod_{j=1}^t q(x_j | x_{j-1})}$$

$$q(x_t | x_0) = \sum_{x_{t-1}} \sum_{x_{t-2}} \dots \sum_{x_1} q(x_t, x_{t-1}, x_{t-2}, \dots, x_1 | x_0)$$

$$= \sum_{x_{t-1}} \dots \sum_{x_1} \prod_{j=1}^t q(x_j | x_{j-1})$$

$$\begin{aligned} \sum_{x_{t-1}} \dots \sum_{x_1} q(x_t | x_{t-1}) q(x_{t-1} | x_{t-2}) \dots q(x_1 | x_0) \\ = \prod_{j=1}^t q(x_j = x_0 | x_{j-1} = x_0) = \prod_{j=1}^t \frac{x_j}{x_{j-1}} = x_t \end{aligned}$$

Reverse process

- For reverse process, we start from all masks, that is $\mathbf{x}_T = \mathbf{m}$ or $P(\mathbf{x}_T) = \delta(\mathbf{x}_T = \mathbf{m})$
- Over subsequent steps, we convert masks into valid words.
- A position once converted from mask to a word, continues to be copied over.

$$P_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t) = P'_{\theta}(x_{t-1,i} | \mathbf{x}_t) \cdot \text{Constraint}(x_{t-1}, x_t) \quad (1)$$

Where,

$$\text{Constraint}(x_{t-1}, x_t) = \delta(x_t, x_{t-1}) \quad \text{if } x_t \neq \mathbf{m} \quad (2)$$

Training objective

Recall, the structured variational objective.

$$\begin{aligned} \max_{\theta, q} \mathbb{E}_{\mathbf{x}_0 \sim D} \mathbb{E}_q [& \log P_{\theta}(\mathbf{x}_0 | \mathbf{x}_1) \\ & + KL(q(\mathbf{x}_T | \mathbf{x}_0); P(\mathbf{x}_T)) \\ & + \sum_{t=1}^{T-1} KL(q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0); P_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1}))] \end{aligned}$$

In the above,

- 1 The first term in the objective is made redundant by making $\mathbf{x}_0 = \mathbf{x}_1$
- 2 The second term also has no parameters to be learned since by design we choose the noise schedule such that $q(\mathbf{x}_T | \mathbf{x}_0) = P(\mathbf{x}_T) = \delta(\mathbf{x}_T = \mathbf{m})$
- 3 We focus on learning only $P_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$, the denoiser via the last KL divergence. Let $L_{t-1} = KL(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0); P_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t))$

Training

First we need to calculate $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$.

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{\sum_{\mathbf{x}} q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)}$$

Substitute in the above:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \text{cat}(\mathbf{x}_t; \frac{\alpha_t}{\alpha_{t-1}}\mathbf{x}_{t-1} + (1 - \frac{\alpha_t}{\alpha_{t-1}})\mathbf{m})$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_0) = \text{cat}(\mathbf{x}_{t-1}; \alpha_{t-1}\mathbf{x}_0 + (1 - \alpha_{t-1})\mathbf{m})$$

And it can be shown that

$$\begin{aligned} q(x_{t-1}|x_t, x_0) &= \frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t} \\ &= \frac{1 - \alpha_{t-1}}{1 - \alpha_t} \\ &= \text{cat}(x_{t-1}; x_t) \end{aligned}$$

$$\text{if } x_t = \mathbf{m}, x_{t-1} = x_0$$

$$\text{if } x_t = \mathbf{m}, x_{t-1} = \mathbf{m}$$

$$\text{if } x_t \neq \mathbf{m}$$

Proof

First, consider $x_t = m$

$$q(x_{t-1} = x_0 | x_t = m) \propto \frac{q(x_t = m | x_{t-1} = x_0) q(x_{t+1} = x_0 | x_0)}{= \left(1 - \frac{\alpha_t}{\alpha_{t-1}}\right) \alpha_{t-1} = \frac{\alpha_{t-1} - \alpha_t}{=}}$$

$$q(x_{t-1} = m | x_t = m) \propto \frac{q(x_t = m | x_{t-1} = m) q(x_{t-1} = m | x_0)}{= \frac{(1 - \alpha_{t-1}) - \frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t}}{=}}$$

$$q(x_{t-1} = m | x_t = m) = \frac{1 - \alpha_{t-1}}{1 - \alpha_t} \quad q$$

Consider $x_t \neq m$

$$\frac{q(x_{t-1} = m | x_t \neq m)}{q(x_{t-1} \neq m | x_t \neq m)} \propto \frac{0}{1}$$

$$\frac{q(x_{t-1} \neq m | x_t \neq m)}{q(x_{t-1} \neq m | x_t \neq m)} \propto \frac{1}{1}$$

$$q(x_t | x_{t-1} = m) = 1 \text{ if } x_t = m$$

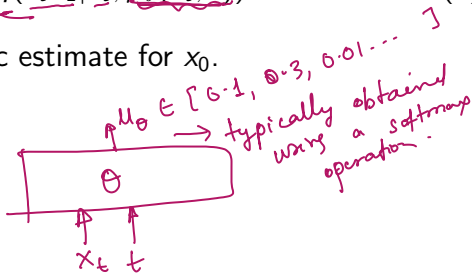
Parameterizing P_θ

For the reverse process, we use

$$p_\theta(x_{t-1}|x_t) = q(x_{t-1}|x_t, \mu_\theta(x_t, t)) \quad (3)$$

where, $\mu_\theta(x_t, t)$ is the parametric estimate for x_0 .

$\mathcal{V}$
 $|V| = R$
 $\mu_\theta \in \Delta^R$



Training P_θ

Expanding the KL divergence term of the training objective, we get:

$$\text{KL}(q(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t))$$

$$= \sum_{x_t \sim q(x_t|x_0)} \sum_{x_{t-1}} q(x_{t-1}|x_t, x_0) \log \frac{q(x_{t-1}|x_t, x_0)}{q(x_{t-1}|x_t, \mu_\theta)}$$

three

We need to consider *three* possible cases for each of x_t, x_{t-1} being the mask token or x_0 .

① $x_t = x_{t-1} \neq m$

$$\text{① } q(x_{t-1} = x_0 | x_t = x_0, x_0) \log \frac{q(x_{t-1} = x_0 | x_t = x_0, x_0)}{q(x_{t-1} = x_0 | x_t = x_0, \mu_\theta)} = 1$$

② $x_t = x_{t-1} = m$

$$\text{② } q(x_{t-1} = m | x_t = m, x_0) = q(x_{t-1} = m | x_t = m, \mu_\theta)$$

KL-term = 0

From the definition of $q(x_{t-1}|x_t, x_0)$, it is clear that the only case where $q(x_{t-1}|x_t, x_0) \neq q(x_{t-1}|x_t, \mu_\theta)$ is when $x_{t-1} \neq \mathbf{m}, x_t = \mathbf{m}$. In this case

$$q(x_{t-1}|x_t, x_0) = \frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t} \quad \left| \begin{array}{l} \delta(x_{t-1}, x_0) \\ \text{if } x_t = x_0 \end{array} \right.$$

$$q(x_{t-1}|x_t, \mu_\theta) = \frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t} \quad \left| \begin{array}{l} \mu_\theta[x_0] \\ \text{if } x_t = \mathbf{m} \end{array} \right.$$

$P(x_{t-1} = x_0 | x_t = \mathbf{m})$

Overall training objective

$$\text{KL}(q(x_{t-1}|x_t, x_0) || q(x_{t-1}|x_t, \mu_\theta(x_t, t))) \quad (4)$$

$$= \frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t} \log \frac{\frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t}}{\frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t} \mu_\theta(x_t, t)[x_0]} \quad (5)$$

$$= \frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t} \log \mu_\theta(x_t, t)[x_0] \delta(x_t = \text{mask}) \quad (6)$$

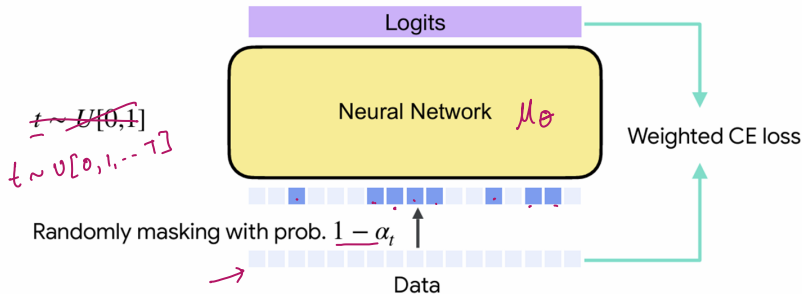
$= x_0^T \log u_\theta(x_t, t)$

Training details

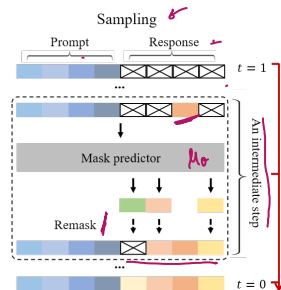
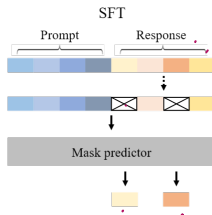
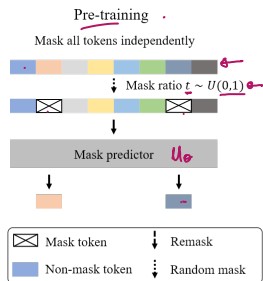
Continuous-time Negative ELBO ($T \rightarrow \infty$)

$$\mathcal{L}_\infty = \int_0^1 \frac{\alpha'_t}{1 - \alpha_t} \mathbb{E}_{q(x_t|x_0)} [\delta_{x_t,m} \cdot x_0^\top \log \mu_\theta(x_t, t)] dt$$

Handwritten notes: $\sum_{t \sim U[0,1] \dots T}$ above the integral; $d_{t-1} - d_t$ next to the integral; $\alpha'_t =$ and $1 - \alpha_t$ circled; $\delta_{x_t,m}$ circled; $x_0^\top \log \mu_\theta(x_t, t)$ underlined.



Large Language Diffusion Models LLaDA²



²<https://ml-gsai.github.io/LLaDA-demo/>

Inference in Llada

- 0: Input: Model: M_θ , Prompt: P , n : number of tokens to generate, T : maximum number of steps.
- 0: Initialize $\mathbf{x}_T = n$ masks
- 0: **for** $t = T$ to 1 **do**
- 0: Invoke $M_\theta(P, \mathbf{x}_t)$, obtain $\mu_i(\cdot)$ for all masked positions i in $\mathbf{x}_t$
- 0: $\mathbf{x}_{t-1}^\sim =$ sample high probability tokens from $\mu_i(\cdot)$ for each i .
- 0: Remask $\frac{1-\alpha_{t-1}}{1-t}$ fraction of tokens in $\mathbf{x}_{t-1}^\sim$. Several heuristics for remasking: random, lowest confidence, block auto-regressive.
- 0: **end for**

Pros and Cons of Diffusion Models

- ① Pros: State of the art accuracy for image generation tasks
- ② Forms the basis of many impressive image generation models e.g. Stable diffusion, Imagen, etc.
- ③ Limitation: Slow multi-step generation, unlike in GANs and VAEs where generation happens in one step.

Future work

- 1 In spite of impressive demos on text-to-image generation, fine-grained control of the generation process still not well-understood.
- 2 Diffusion models have not been successful on discrete data such as language.